

Verilator lint

Lint is cheap insurance

Common warning families

- Watch for UNUSED, something you declared but never read
- WIDTH, expressions that do not line up
- LATCH, inferred storage you did not mean
- CASEINCOMPLETE, case statements without a default or full coverage
- The literacy lab tags these families so you learn the fix, not only the flag name

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Verilator lint lab

Document-aid literacy: **-Wall** warning categories (UNUSED, WIDTH, LATCH, ...) — not a real Verilator run. Starter: **-Wall** on · findings fixed — CLEAN.

Starter example: **-Wall** on · all findings fixed — board CLEAN.

Load starter example

Challenges 0 / 22 cleared

Quiz: **-Wall:** **-Wall** is used to...

☐ enable a useful set of lint warning categories

☐ wipe the VCD file

☐ select UVM_TESTNAME

☐ force --trace off

Show hint

Check

Next

Idle

1

2

3

4

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6

7

8

9

10

11

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Core ideas

-Wall

Enable useful lint warning categories.

UNUSED / WIDTH

Dead signals and size mismatches.

-Werror

Promote remaining warns to CI failures.

-Wno-*

Silence one code — prefer fixing RTL.

Lab

SCENARIO

clean starter

Load preset

Fix selected

Scan lint

Demo UNUSED

Explain

Reset

Lab starter

Real Verilator practice

- In Track A, lint a tiny module with wall-level warnings enabled
- Introduce one intentional WIDTH or UNUSED issue
- Avoid the reflex to blanket-disable warnings unless you have documented why
- Self-check scripts can confirm you reached a clean lint pass

Pitfalls to watch

- Do not blanket-disable warnings to greenwash a broken design
- Do not confuse BLIND with CLEAN, no output is not the same as no problems
- Do not ship with warnings-as-errors until your tree is actually clean
- And remember: lint passes do not replace simulation, they narrow the search space

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, reach CLEAN on at least one challenge without only silencing flags
- In Track A, run lint with strict warnings once
- When you are ready, take the short quiz, then continue to C++ testbench and DPI